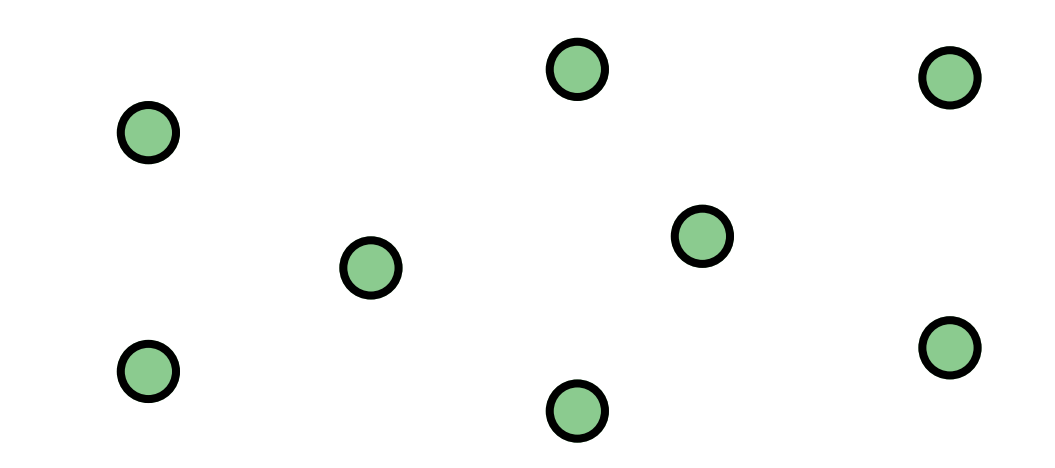


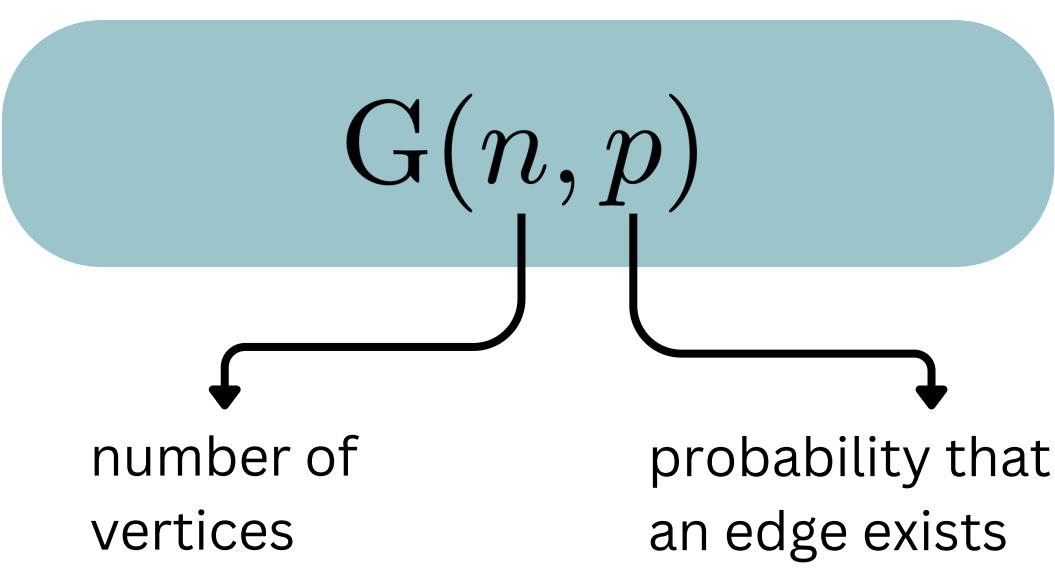
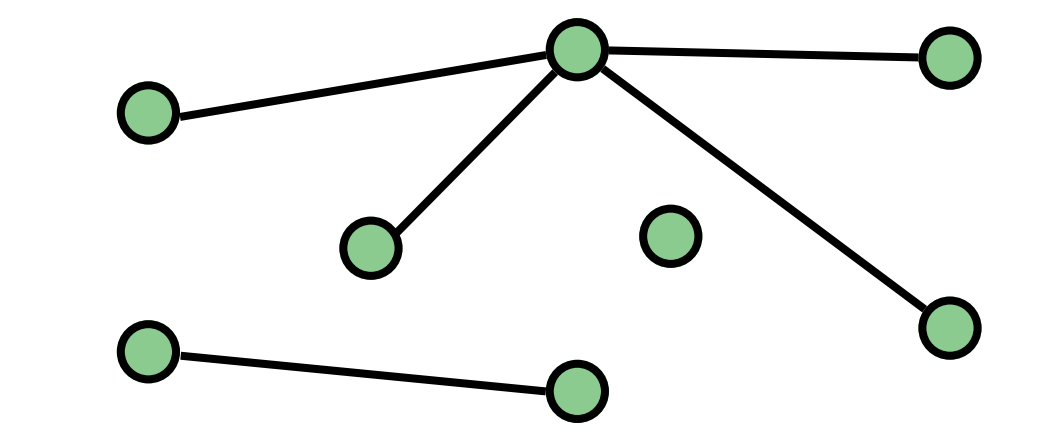
A modern look into the Erdős–Rényi random process

Erdős–Rényi random graph

1 Start with n isolated vertices



2 For each pair of vertices, add an edge with probability p , independently



What does it model?

Epidemics
spreading of a disease according to connectivity among individuals

Networks
connection among websites according to links

Percolation
possibility of a fluid to flow through a material

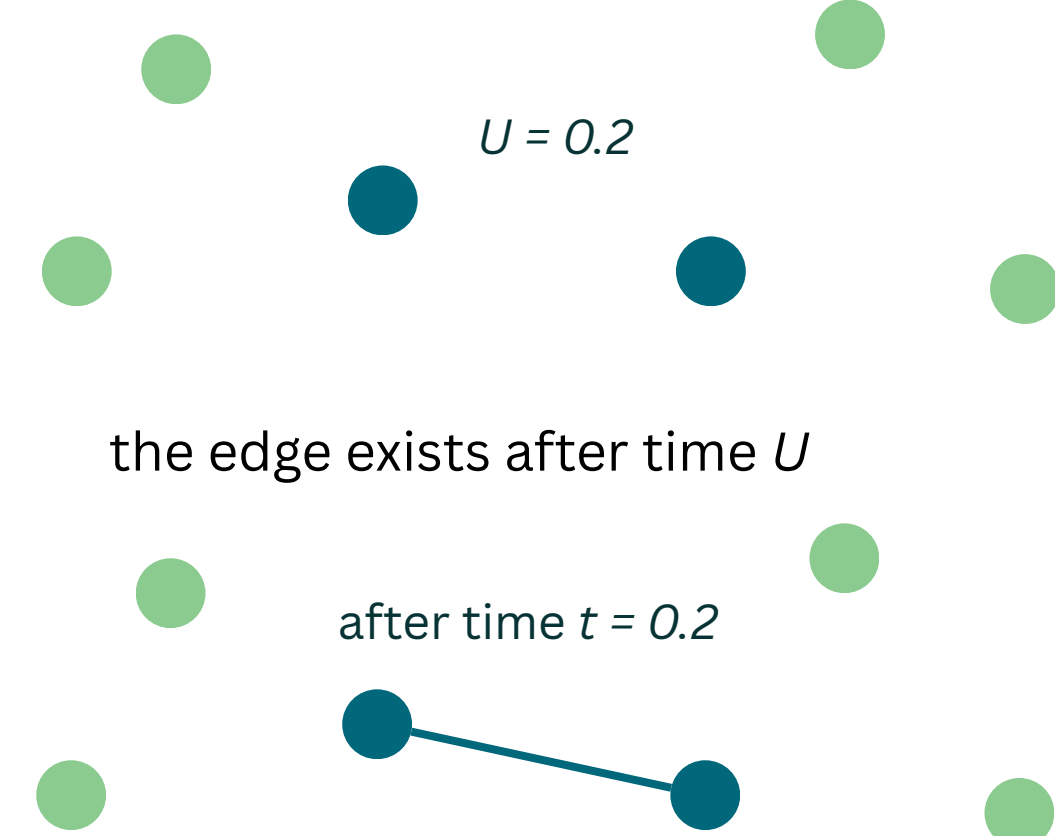
Warning: this model is limited for applications

- there is no geometry
- edges exist with the same probability and independently

For n fixed, there is a natural way to embed every Erdős–Rényi random graph in the same space

1 Start with n isolated vertices

2 For each pair of vertices pick a random number U in $[0,1]$



now, instead of one random graph, we have an Erdős–Rényi random graph with probability t for every t in $[0,1]$

$$(G(n, t))_{t \in [0,1]}$$

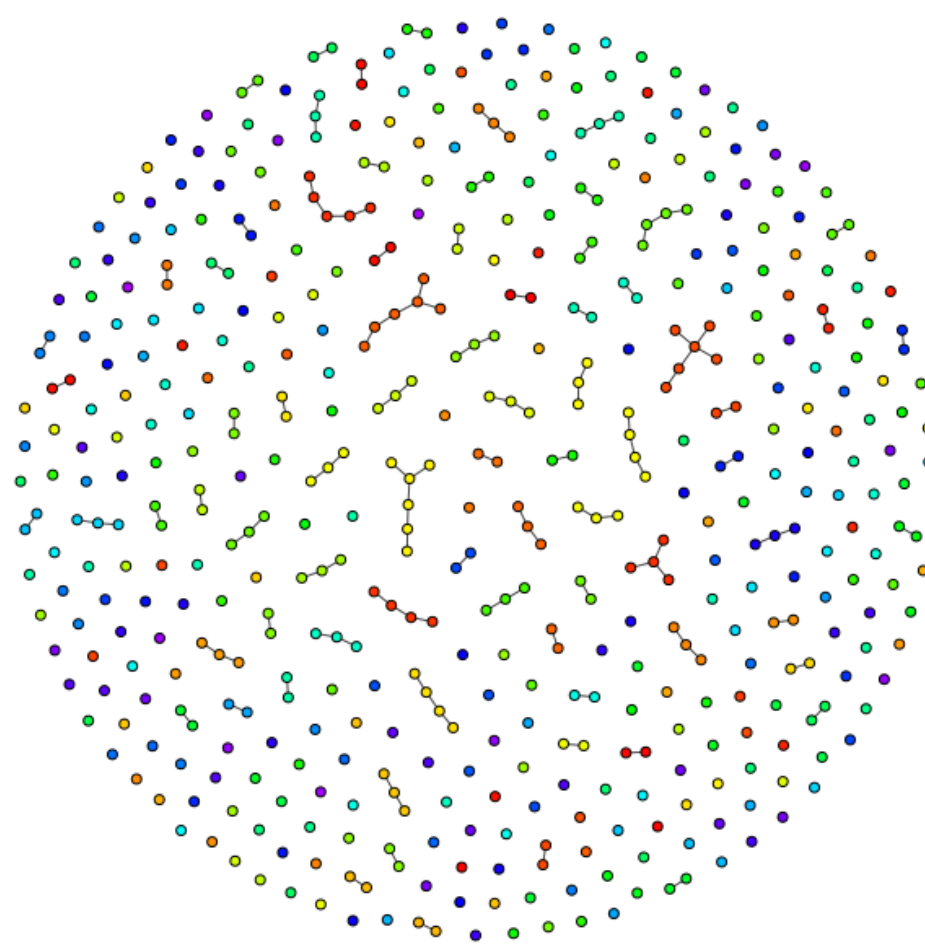
The phase transition

We will study the behavior of $(G(n, \frac{c}{n}))_{c \geq 0}$

THE MEAN NUMBER OF EDGES FROM ANY VERTEX IS c

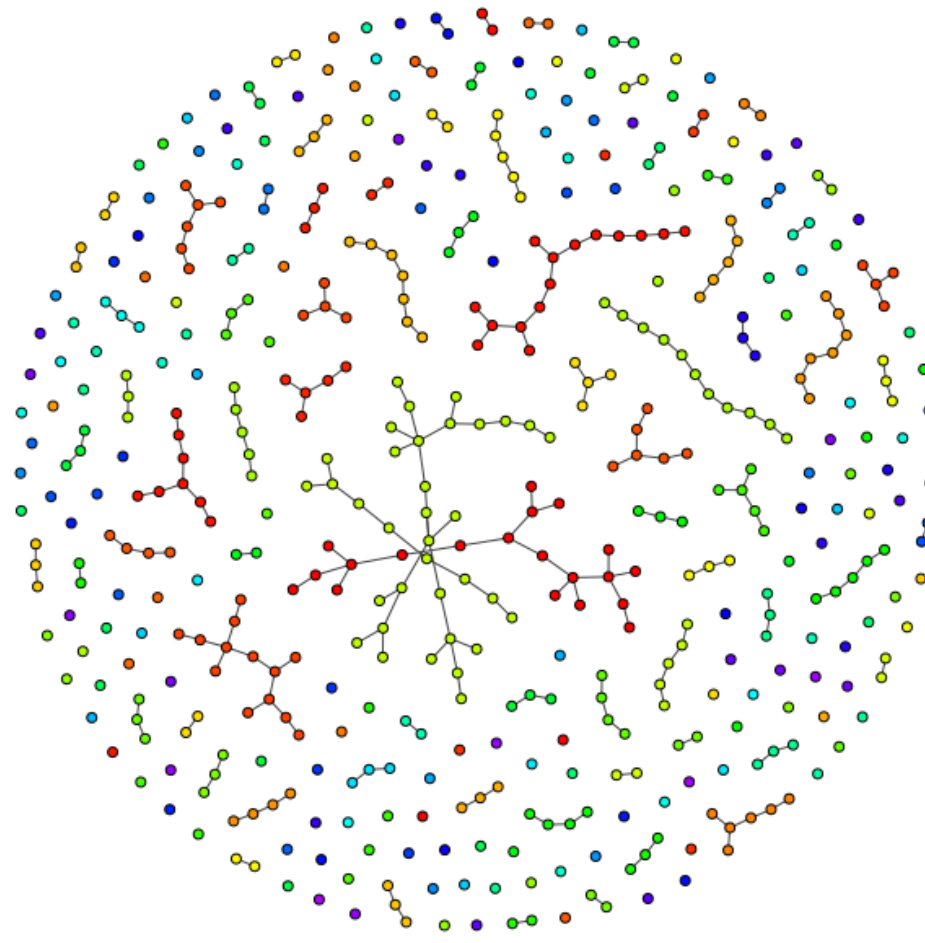
1. sub-critical $0 \leq c < 1$

few edges
connected components are small
infection stays in small groups: there is no epidemic



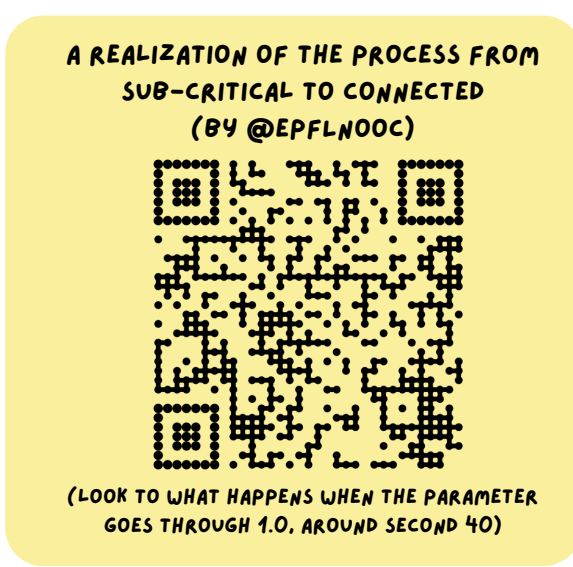
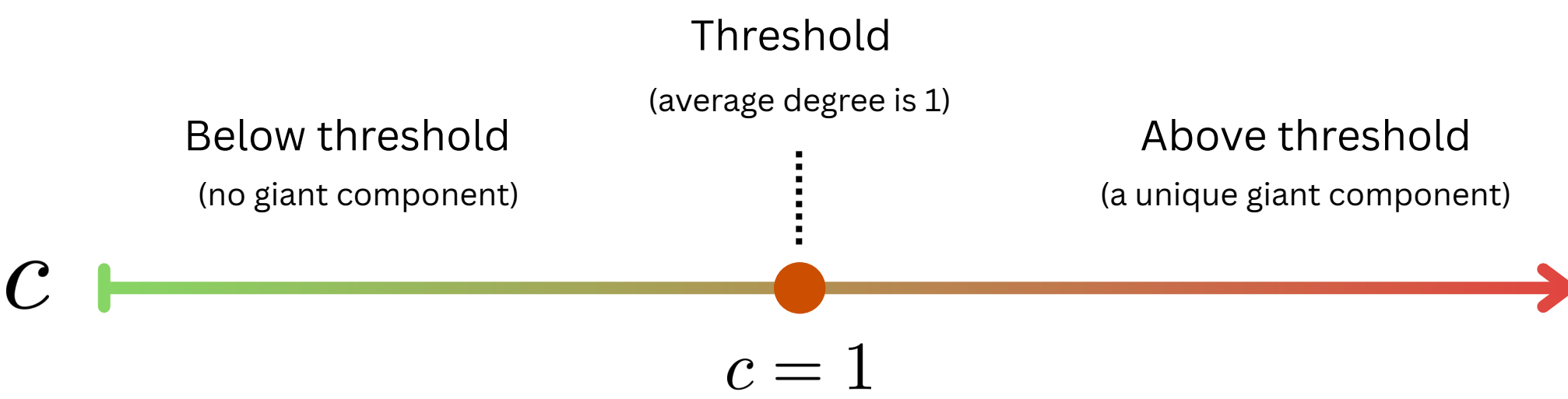
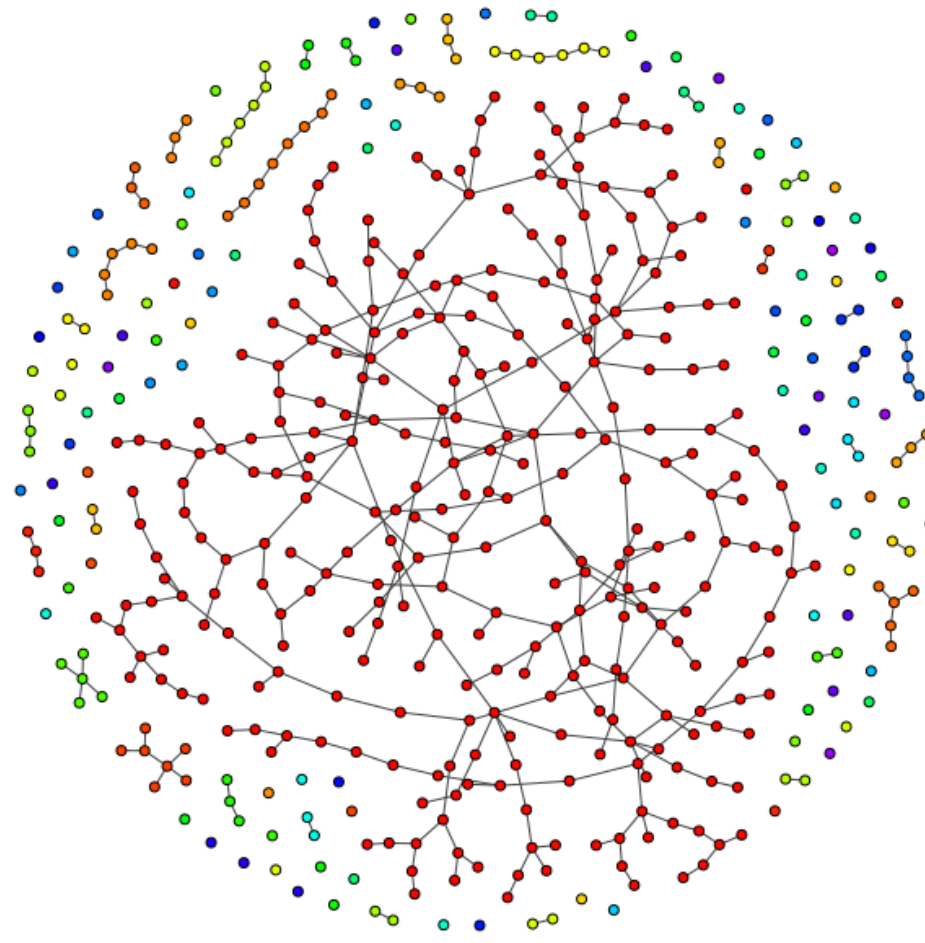
2. critical $c = 1$

average degree of a vertex is 1
many connected components of medium size
a giant component is about to appear



3. super-critical $c > 1$

average degree of a vertex is finite but larger than 1
there is a unique giant component
infection can propagate through larger groups: epidemic



Number of connected components

$$K\left(\frac{c}{n}\right) \longrightarrow \text{number of connected components in } G\left(n, \frac{c}{n}\right)$$

Puhalskii (2005):

Fluid limit: for every $0 \leq c < 1$

$$\frac{1}{n} K\left(\frac{c}{n}\right) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 1 - \frac{c}{2}$$

Diffusion limit: for every $0 \leq c < 1$

$$\sqrt{n} \left(\frac{1}{n} K\left(\frac{c}{n}\right) - \left(1 - \frac{c}{2}\right) \right) \xrightarrow[n \rightarrow \infty]{\text{law}} N\left(0, \frac{c}{2}\right)$$

GAUSSIAN DISTRIBUTION WITH MEAN 0 AND VARIANCE $c/2$ (THE SAME DISTRIBUTION AS IN THE CENTRAL-LIMIT THEOREM)

Size of the giant component

$$L\left(\frac{c}{n}\right) \longrightarrow \text{size of the largest connected components in } G\left(n, \frac{c}{n}\right)$$

Erdős and Rényi (1959), Stepanov (1970):

Fluid limit: for every $0 \leq c < 1$

$$\frac{1}{n} L\left(\frac{c}{n}\right) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \rho(c)$$

Diffusion limit: for every $0 \leq c < 1$

$$\sqrt{n} \left(\frac{1}{n} L\left(\frac{c}{n}\right) - \rho(c) \right) \xrightarrow[n \rightarrow \infty]{\text{law}} N\left(0, \sigma^2(c)\right)$$

$$\text{where } \sigma^2(c) = \frac{\rho(c)(1 - \rho(c))}{(1 - c(1 - \rho(c)))^2}$$

what can we say about the limit of the processes of fluctuations

$$\left(\sqrt{n} \left(\frac{L(c/n)}{n} - \rho(c) \right), c > 1 \right) \text{ and } \left(\sqrt{n} \left(\frac{1}{n} K\left(\frac{c}{n}\right) - \left(1 - \frac{c}{2}\right) \right), c < 1 \right)$$

what is the limit of

$$\text{Cov} \left(L\left(\frac{c_1}{n}\right), L\left(\frac{c_2}{n}\right) \right) \text{ and } \text{Cov} \left(K\left(\frac{c_1}{n}\right), K\left(\frac{c_2}{n}\right) \right)$$

Dynamical fluctuations

C. (2024+)

$$\left(\sqrt{n} \left(\frac{K(c/n)}{n} - \left(1 - \frac{c}{2}\right) \right), c < 1 \right)$$

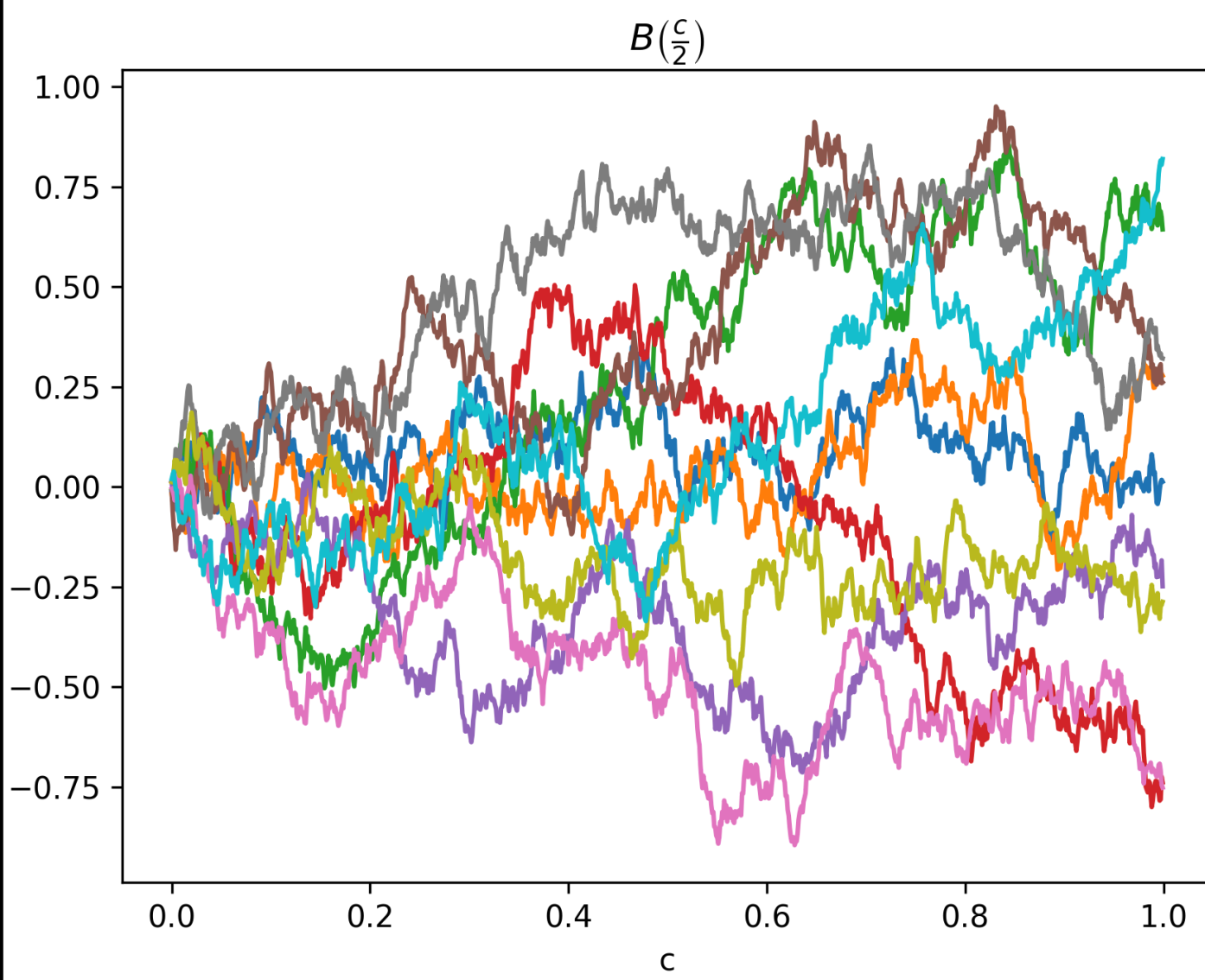
converges in law to

$$\left(B\left(\frac{c}{2}\right), c < 1 \right)$$

standard Brownian motion

$$\frac{1}{n} \text{Cov} \left(K\left(\frac{c_1}{n}\right), K\left(\frac{c_2}{n}\right) \right) \xrightarrow[n \rightarrow \infty]{} \frac{c_1}{2}$$

for $0 \leq c_1 \leq c_2 < 1$



Remarks

- the result is actually stated for more general (inhomogeneous) random graph processes
- the proof consists on the study of a martingale related to the multiplicative coalescent dynamics

**Enriquez, Faraud and Lemaire (2023)
C., Lemaire and Limic (2024+)**

$$\left(\sqrt{n} \left(\frac{L(c/n)}{n} - \rho(c) \right), c > 1 \right)$$

converges in law to

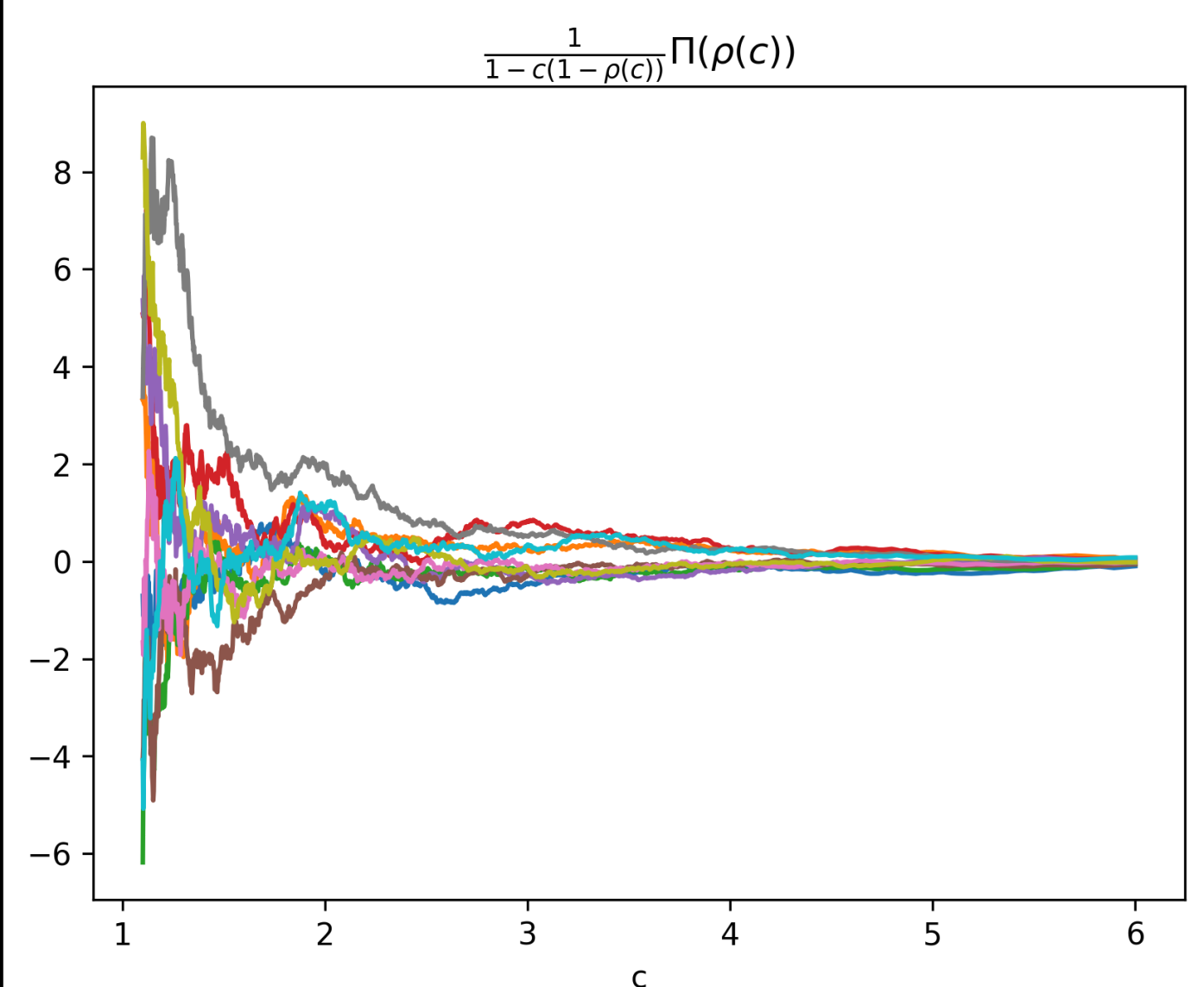
$$\left(\frac{1}{1 - c(1 - \rho(c))} \Pi(\rho(c)), c > 1 \right)$$

STANDARD BROWNIAN BRIDGE

$$\frac{1}{n} \text{Cov} \left(L\left(\frac{c_1}{n}\right), L\left(\frac{c_2}{n}\right) \right) \text{ converges to}$$

$$\frac{\rho(c_1)(1 - \rho(c_2))}{(1 - c_1(1 - \rho(c_1)))(1 - c_2(1 - \rho(c_2)))}$$

for $0 \leq c_1 \leq c_2 < 1$



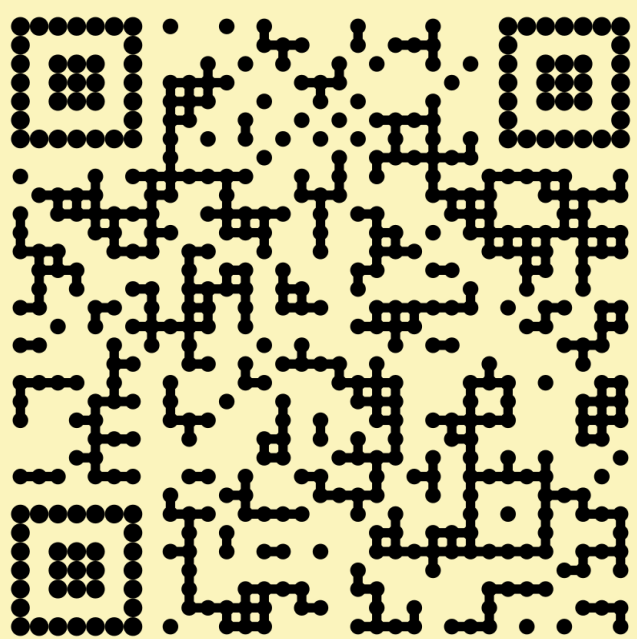
Remarks

- our proof is based on the study of an encoding process called simultaneous breadth-first walk, introduced by Limic (2019)
- in the same article, they also study the fluctuations in the barely super-critical regime, when the time is $\frac{1+t \cdot \epsilon_n}{n}$ with $\epsilon_n \rightarrow 0$ and $\sqrt[3]{n} \cdot \epsilon_n \rightarrow \infty$
- by using the same techniques, Clancy Jr. (2025), proves an analogous result for a class of inhomogeneous random graph processes

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